

GCSE Mathematics - Foundation Tier

Paper 3 (Calculator) Practice - Set 9

Time: 1 hour 30 minutes | Total marks: 80 | Calculator allowed

Instructions

- Use black ink or ball-point pen.
- Answer ALL questions in the spaces provided.
- You must show all your working - answers given without working may not score full marks.
- Diagrams are NOT accurately drawn unless otherwise stated.
- If your calculator does not have a pi button, use $\pi = 3.142$.

1.

Write $\frac{3}{5}$ as a decimal.

(Total for Question 1 is 1 marks)

2.

Work out 45% of £180.

(Total for Question 2 is 2 marks)

3.

- (a) Write 4^3 as an ordinary number. (1)
- (b) Write $\sqrt{81}$ as an ordinary number. (1)

(Total for Question 3 is 2 marks)

4.

Write the ratio 35 : 49 in its simplest form.

(Total for Question 4 is 2 marks)

5.

The frequency table shows the number of texts sent by 30 students in one day.

Texts	0-9	10-19	20-29	30-39	40-49
Freq.	4	9	11	4	2

- (a) Write down the modal class. (1)
- (b) Work out the percentage of students who sent at least 20 texts. (2)

(Total for Question 5 is 3 marks)

6.

Solve $3(x - 2) = 9$.

(Total for Question 6 is 2 marks)

7.

A right-angled triangle has a hypotenuse of 25 cm and one shorter side of 7 cm.

[Diagram: right-angled triangle with hypotenuse 25 cm and one shorter side 7 cm; the other side labelled x cm.]

Work out the length x .

(Total for Question 7 is 3 marks)

8.

The table shows the number of goals scored per match by a netball team over a season.

Goals	20	25	30	35	40
Freq.	3	5	8	6	3

Work out the mean number of goals scored per match. Give your answer to 2 decimal places.

(Total for Question 8 is 3 marks)

9.

A boat sails from point A on a bearing of 045 deg to point B. Then it sails from B on a bearing of 135 deg to point C.

[Diagram: point A with north line; AB on bearing 045 deg; B with its own north line; BC on bearing 135 deg.]

- (a) Work out the angle ABC. (2)
- (b) Hence describe the triangle ABC. (1)

(Total for Question 9 is 3 marks)

10.

The cost C (in pounds) of running a freezer for d days is given by $C = 0.85d + 12$.

- (a) Work out C when $d = 30$. (2)
- (b) How many days does it take to spend exactly £80? (Solve for d .) (1)

(Total for Question 10 is 3 marks)

11.

A cuboid has a square base of side 4 cm and a height of 9 cm.

- (a) Work out the volume of the cuboid. (1)
- (b) Work out the total surface area of the cuboid. (2)

(Total for Question 11 is 3 marks)

12.

Nadia puts £1500 into a savings account paying 1.5% compound interest per year.

Work out the value of the investment after 4 years. Give your answer to the nearest penny.

(Total for Question 12 is 4 marks)

13.

In a game, a fair coin is tossed and a fair four-sided die is rolled at the same time. The die has faces numbered 1, 2, 3 and 4.

- (a) Write down the total number of possible outcomes. (1)
- (b) Work out the probability of getting heads on the coin AND a 3 on the die. (3)

(Total for Question 13 is 4 marks)

14.

A right-angled triangle has shorter sides of length 9 cm and 12 cm.

[Diagram: right-angled triangle with shorter sides 9 cm and 12 cm; hypotenuse labelled.]

- (a) Show that the hypotenuse is 15 cm. (3)
- (b) Hence work out the perimeter of the triangle. (2)

(Total for Question 14 is 5 marks)

15.

A piece of metal in the shape of a triangular prism has the following dimensions: cross-section is a right-angled triangle with shorter sides 5 cm and 12 cm; length of prism is 25 cm.

[Diagram: right-angled triangular prism: cross-section sides 5 cm and 12 cm; length 25 cm.]

- (a) Work out the volume of the prism. (3)

The metal has a density of 4.5 g/cm^3 .

- (b) Work out the mass of the prism. Give your answer in kilograms. (2)

(Total for Question 15 is 5 marks)

16.

The first four terms of an arithmetic sequence are 4, 9, 14, 19.

- (a) Find an expression for the n th term. (2)
- (b) Work out the 30th term of the sequence. (2)

(Total for Question 16 is 4 marks)

17.

The table shows the age (years) and resting heart rate (bpm) of 8 adults.

Age	20	25	30	35	40	45	50	55
Rate	62	65	68	70	72	75	77	80

- (a) On the grid provided, plot these data as a scatter graph. (2)

- (b) Describe the correlation. (1)
- (c) Draw a line of best fit. (1)
- (d) Use your line of best fit to estimate the resting heart rate of a 33 year old. (1)

(Total for Question 17 is 5 marks)

18.

A cylindrical water tank has a radius of 60 cm and a height of 1 m.

- (a) Show that the volume of the tank is approximately 1.131 m^3 (use $\pi = 3.142$). (4)

The tank is being filled at a rate of 30 litres per minute (and $1 \text{ m}^3 = 1000$ litres).

- (b) Work out the time, in minutes (to the nearest minute), to fill the tank. (1)

(Total for Question 18 is 5 marks)

19.

A scale drawing uses 1 cm to represent 5 km.

- (a) A road is 8.5 cm long on the drawing. Work out the real length in kilometres. (2)

Two towns P and Q are 32.5 km apart in real life.

- (b) Work out the distance between P and Q on the scale drawing. (2)
- (c) Express the scale of the drawing in the form $1 : n$. (1)

(Total for Question 19 is 5 marks)

20.

The sum of the first n terms of an arithmetic sequence is given by $S_n = n^2 + 2n$.

- (a) Work out S_3 (the sum of the first three terms). (1)
- (b) Work out S_4 (the sum of the first four terms). (1)
- (c) Hence work out the 4th term of the sequence. (1)
- (d) Show that the sequence has first term 3. (2)

(Total for Question 20 is 5 marks)

21.

A bank offers two savings accounts.

Account X: 2.5% compound interest per year.

Account Y: 2.8% simple interest per year.

Sam invests £4000 in each.

- (a) Work out the value of Account X after 5 years. Give your answer to the nearest penny. (3)
- (b) Work out the value of Account Y after 5 years. (2)
- (c) Which account is worth more after 5 years? By how much? (1)

(Total for Question 21 is 6 marks)

22.

There are 40 chocolates in a box: 18 milk chocolates, 14 dark chocolates and 8 white chocolates.

Tom takes a chocolate at random, notes its type, then puts it back. He then takes a second chocolate at random.

- (a) Write down the probability that a single chocolate taken at random is milk. Give your answer in its simplest form. (1)
- (b) Work out the probability that BOTH chocolates taken are milk. (2)
- (c) Work out the probability that NEITHER chocolate taken is milk. (2)

(Total for Question 22 is 5 marks)

TOTAL FOR PAPER: 80 MARKS

Mark Scheme

Foundation Tier - Paper 3 (Calculator) - Set 9 | 80 marks

Question 1

- 0.6 - B1

Total: 1 marks

Question 2

- 180×0.45 or $180 \times 45 / 100$ - M1
- £81 - A1

Total: 2 marks

Question 3

- (a) 64 - B1
- (b) 9 - B1

Total: 2 marks

Question 4

- Both parts divided by 7 ($35:49 \rightarrow 5:7$) - M1
- 5 : 7 - A1

Total: 2 marks

Question 5

- (a) 20 - 29 - B1
- (b) $11 + 4 + 2 = 17$ students; $17/30 \times 100$ - M1
- (b) 56.7% (accept 56.6 - 56.7 to 1 dp) - A1

Total: 3 marks

Question 6

- Divide by 3 ($x - 2 = 3$) OR expand ($3x - 6 = 9$) - M1
- $x = 5$ - A1

Total: 2 marks

Question 7

- $x^2 = 25^2 - 7^2$ - M1
- $625 - 49 = 576$ - M1
- $x = 24$ (cm) - A1

Total: 3 marks

Question 8

- $\text{Sum}(fx) = 60 + 125 + 240 + 210 + 120 = 755$ - M1
- $\text{Sum}(f) = 25$ and $755 / 25$ - M1
- 30.20 - A1

Total: 3 marks

Question 9

- (a) Angle from bearing AB (045) plus turn at B to bearing BC (135): supplementary angle to consider - M1
- (a) 90 (deg) - A1
- (b) Right-angled triangle - B1

Total: 3 marks

Question 10

- (a) $0.85 \times 30 + 12 = 25.50 + 12$ - M1
- (a) £37.50 - A1
- (b) $80 - 12 = 68$; $/ 0.85 = 80$ days - B1

Total: 3 marks

Question 11

- (a) $4 \times 4 \times 9 = 144$ (cm³) - B1
- (b) $2(4 \times 4) + 4(4 \times 9) = 32 + 144$ - M1
- (b) 176 (cm²) - A1

Total: 3 marks

Question 12

- Multiplier 1.015 seen - M1
- $1500 \times 1.015^2 (= 1545.3375)$ seen - M1
- 1500×1.015^4 - M1
- £1592.04 (accept £1592.045) - A1

Total: 4 marks

Question 13

- (a) 8 (= 2×4) - B1
- (b) $P(\text{heads}) = 1/2$ and $P(3) = 1/4$ identified - M1
- (b) $1/2 \times 1/4$ - M1
- (b) $1/8$ - A1

Total: 4 marks

Question 14

- (a) $c^2 = 9^2 + 12^2$ - M1
- (a) $81 + 144 = 225$ - M1
- (a) $c = \sqrt{225} = 15$ (cm) shown - C1
- (b) $9 + 12 + 15$ - M1
- (b) 36 (cm) - A1

Total: 5 marks

Question 15

- (a) $0.5 \times 5 \times 12 = 30$ (area of cross-section) - M1
- (a) 30×25 - M1
- (a) 750 (cm³) - A1
- (b) $4.5 \times 750 = 3375$ (g) - M1

- (b) 3.375 (kg) - A1

Total: 5 marks

Question 16

- (a) Common difference 5 noted; zeroth term -1 - M1
- (a) $5n - 1$ - A1
- (b) $5 \times 30 - 1$ - M1
- (b) 149 - A1

Total: 4 marks

Question 17

- (a) 8 points plotted correctly - B2 (B1 for 6+ correct)
- (b) Positive correlation - B1
- (c) Sensible line of best fit - B1
- (d) Heart rate at 33 yr (accept 67 - 71 bpm) - B1

Total: 5 marks

Question 18

- (a) Convert: 60 cm = 0.6 m, height 1 m - M1
- (a) $V = \pi \times 0.6^2 \times 1$ (= 3.142 x 0.36) - M1
- (a) 1.13112 - M1
- (a) 1.131 (m³) shown - C1
- (b) $1131 / 30 = 37.7$ minutes; rounded 38 minutes - B1

Total: 5 marks

Question 19

- (a) 8.5×5 - M1
- (a) 42.5 km - A1
- (b) $32.5 / 5$ - M1
- (b) 6.5 cm - A1
- (c) 1 cm = 5 km = 500 000 cm so 1 : 500 000 - B1

Total: 5 marks

Question 20

- (a) $3^2 + 2(3) = 9 + 6 = 15$ - B1
- (b) $4^2 + 2(4) = 16 + 8 = 24$ - B1
- (c) 4th term = $S_4 - S_3 = 24 - 15 = 9$ - B1
- (d) $S_1 = 1 + 2 = 3$ - first term equals S_1 - M1
- (d) Therefore first term is 3 - A1

Total: 5 marks

Question 21

- (a) Multiplier 1.025 seen - M1
- (a) 4000×1.025^4 (= 4415.25) seen - M1
- (a) £4525.63 - A1
- (b) $0.028 \times 4000 \times 5 = 560$ - M1

- (b) $4000 + 560 = £4560$ - A1
- (c) Y is worth more by $4560 - 4525.63 = £34.37$ - B1

Total: 6 marks

Question 22

- (a) $18 / 40 = 9/20$ - B1
- (b) $(9/20) \times (9/20)$ or $(18/40)^2$ - M1
- (b) $81/400$ (accept 0.2025) - A1
- (c) $P(\text{not milk}) = 22/40 = 11/20$; $(11/20)^2$ - M1
- (c) $121/400$ (accept 0.3025) - A1

Total: 5 marks

Worked Solutions

Foundation Tier - Paper 3 (Calculator) - Set 9 | Detailed explanations

Question 1

$$3/5 = 3 \div 5 = 0.6.$$

Question 2

$$45\% \text{ of } 180 = 0.45 \times 180 = 81.$$

Answer: £81.

Question 3

(a) $4^3 = 4 \times 4 \times 4 = 64.$

(b) $\sqrt{81} = 9$ (because $9 \times 9 = 81$).

Question 4

HCF of 35 and 49 is 7.

$$35/7 = 5, 49/7 = 7. \text{ Answer: } 5 : 7.$$

Question 5

(a) Highest frequency is 11, in the 20-29 group. Modal class: 20-29.

(b) "At least 20" = $11 + 4 + 2 = 17$ students.

$$\text{Percentage} = 17 / 30 \times 100 = 56.66\ldots\%, \text{ i.e. } 56.7\% \text{ to 1 dp.}$$

Question 6

$$3(x - 2) = 9. \text{ Divide by 3: } x - 2 = 3. \text{ Add 2: } x = 5.$$

$$\text{Check: } 3(5 - 2) = 3 \times 3 = 9.$$

Question 7

$$\text{Pythagoras for the shorter side: } x^2 = 25^2 - 7^2 = 625 - 49 = 576.$$

$$x = \sqrt{576} = 24 \text{ cm.}$$

Question 8

$$\text{Sum}(fx) = 20 \times 3 + 25 \times 5 + 30 \times 8 + 35 \times 6 + 40 \times 3 = 60 + 125 + 240 + 210 + 120 = 755.$$

$$\text{Sum}(f) = 3 + 5 + 8 + 6 + 3 = 25.$$

$$\text{Mean} = 755 / 25 = 30.20.$$

Question 9

(a) At B, two bearings meet: the back-bearing of AB at B is $045 + 180 = 225$ deg, and the new bearing BC is 135 deg.

Angle ABC (the interior angle of triangle ABC at B) = $225 - 135 = 90$ deg.

(b) Since angle ABC = 90 deg, triangle ABC is a right-angled triangle (right angle at B).

Question 10

(a) $C = 0.85 \times 30 + 12 = 25.50 + 12 = 37.50$.

Answer: £37.50.

(b) Solve $0.85d + 12 = 80$. So $0.85d = 68$ and $d = 80$ days.

Question 11

(a) $V = 4 \times 4 \times 9 = 144 \text{ cm}^3$.

(b) Top and bottom: $2(4 \times 4) = 32$. Four side rectangles: $4(4 \times 9) = 144$. Total = 176 cm^2 .

Question 12

Multiplier 1.015.

After year 1: $1500 \times 1.015 = \text{£}1522.50$.

Year 2: $1522.50 \times 1.015 = \text{£}1545.3375$.

Year 3: $1545.3375 \times 1.015 = \text{£}1568.5176\ldots$

Year 4: $1568.5176\ldots \times 1.015 = \text{£}1592.0454\ldots$

Or $1500 \times 1.015^4 = 1500 \times 1.061363\ldots = \text{£}1592.0454\ldots$

To the nearest penny: £1592.04.

Question 13

(a) 2 outcomes for coin $\times$ 4 for die = 8.

(b) Coin and die are independent.

$P(\text{heads}) = 1/2$. $P(\text{die shows } 3) = 1/4$.

$P(\text{both}) = 1/2 \times 1/4 = 1/8$.

Question 14

(a) $c^2 = 9^2 + 12^2 = 81 + 144 = 225$.

$c = \sqrt{225} = 15$. (Shown.)

(b) Perimeter = $9 + 12 + 15 = 36 \text{ cm}$.

Question 15

(a) Area of cross-section = $0.5 \times 5 \times 12 = 30 \text{ cm}^2$.

Volume = $30 \times 25 = 750 \text{ cm}^3$.

(b) Mass = $4.5 \times 750 = 3375 \text{ g} = 3.375 \text{ kg}$.

Question 16

- (a) Common difference 5. Zeroth term = $4 - 5 = -1$. nth term = $5n - 1$.
(b) 30th term = $5(30) - 1 = 149$.
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Question 17

- (a) Plot 8 points.
(b) Positive correlation.
(c) Best fit drawn.
(d) At 33 years, line gives roughly 67 - 71 bpm.
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Question 18

- (a) Radius 60 cm = 0.6 m; height 1 m.
 $V = \pi \times r^2 \times h = 3.142 \times 0.6^2 \times 1 = 3.142 \times 0.36 = 1.13112$.
To 3 dp: 1.131 m^3 . (Shown.)
(b) $1.131 \text{ m}^3 = 1131 \text{ litres}$. Time = $1131 / 30 = 37.7$ minutes, i.e. 38 minutes.
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Question 19

- (a) $8.5 \text{ cm} \times 5 \text{ km/cm} = 42.5 \text{ km}$.
(b) $32.5 \text{ km} / 5 \text{ km/cm} = 6.5 \text{ cm}$.
(c) $5 \text{ km} = 500\,000 \text{ cm}$. So 1 cm represents 500 000 cm, i.e. scale is 1 : 500 000.
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Question 20

- (a) $S_3 = 3^2 + 2(3) = 9 + 6 = 15$.
(b) $S_4 = 4^2 + 2(4) = 16 + 8 = 24$.
(c) 4th term = $S_4 - S_3 = 24 - 15 = 9$.
(d) The first term equals $S_1 = 1^2 + 2(1) = 1 + 2 = 3$.
So the first term is 3.
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Question 21

- (a) Account X (compound): 4000×1.025^5 .
 $1.025^2 = 1.050625$; $1.025^4 = 1.103813\dots$; $1.025^5 = 1.131408\dots$
Value = $4000 \times 1.131408\dots = \text{£}4525.63$.
(b) Account Y (simple): interest per year = $0.028 \times 4000 = \text{£}112$. Over 5 years: $112 \times 5 = \text{£}560$.
Total = $4000 + 560 = \text{£}4560$.
(c) Y is worth more by $4560 - 4525.63 = \text{£}34.37$.
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Question 22

- (a) $P(\text{milk}) = 18 / 40 = 9/20$.
(b) With replacement, the two picks are independent.
 $P(\text{both milk}) = (9/20) \times (9/20) = 81/400 = 0.2025$.

(c) $P(\text{not milk}) = 22 / 40 = 11/20$.

$P(\text{both NOT milk}) = (11/20) \times (11/20) = 121/400 = 0.3025$.
